NEID Assignment 1

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Problem 1

Part (A)

Surface area: $A = \pi d^2$, with $d = 5.64 \times 10^{-4}$ cm:

$$A \approx \pi(5.64 \times 10^{-4})^2 \approx 9.993 \times 10^{-7} \text{ cm}^2.$$

Total capacitance:

$$C_A = C_m A \approx (1 \times 10^{-6})(9.993 \times 10^{-7}) \approx 9.993 \times 10^{-13} \text{ F} \approx 1.0 \text{ pF}.$$

Charge injected in 1 ms:

$$Q = IT = (1 \times 10^{-9})(1 \times 10^{-3}) = 1 \times 10^{-12} \text{ C} = 1 \text{ pC}.$$

Voltage change:

$$\Delta V = \frac{Q}{C_A} \approx \frac{1 \times 10^{-12}}{9.993 \times 10^{-13}} \approx 1.00 \text{ V}.$$

$$V_{m,\text{after pulse}} \approx -70 \text{ mV} + 1000 \text{ mV} \approx 930 \, \text{mV}.$$

Therefore, the final membrane potential after the pulse for Neuron A is approximately 930 mV.

Part (B)

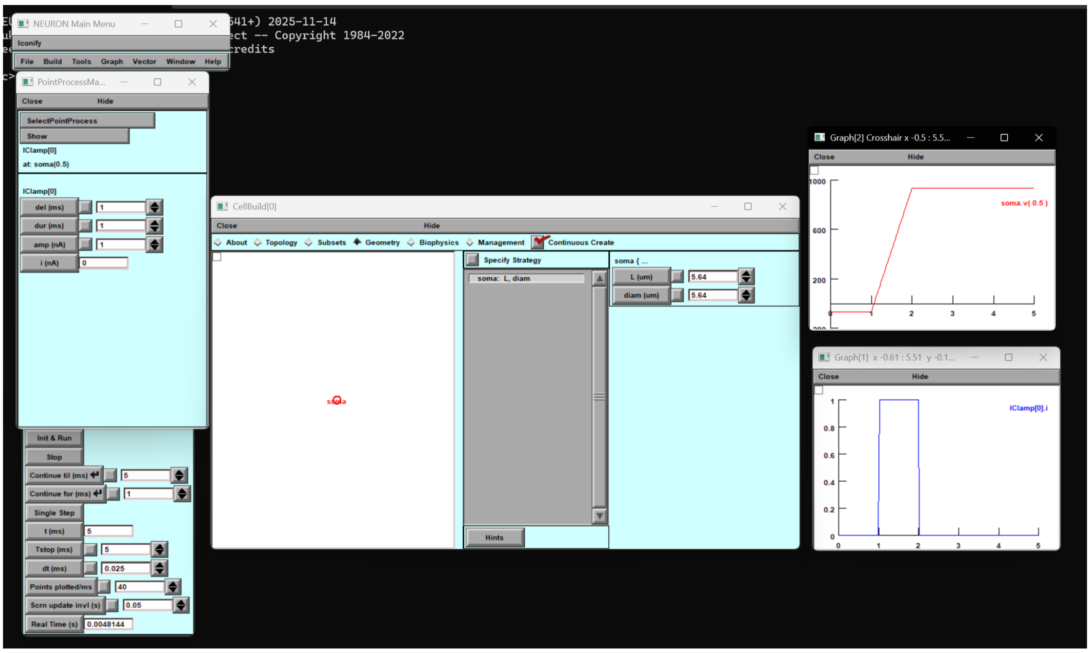

Figure 1: Question 1 Part (b)

Part (C)

Goal: $V_{m,\text{after pulse}} = 50$ mV. Trial-and-error in simulation gave:

$$I_{\text{sim}} = 0.119\,92\,\text{nA}.$$

Therefore, the current amplitude required in simulation to achieve the target final membrane potential of 50 mV was 0.119 92 nA.

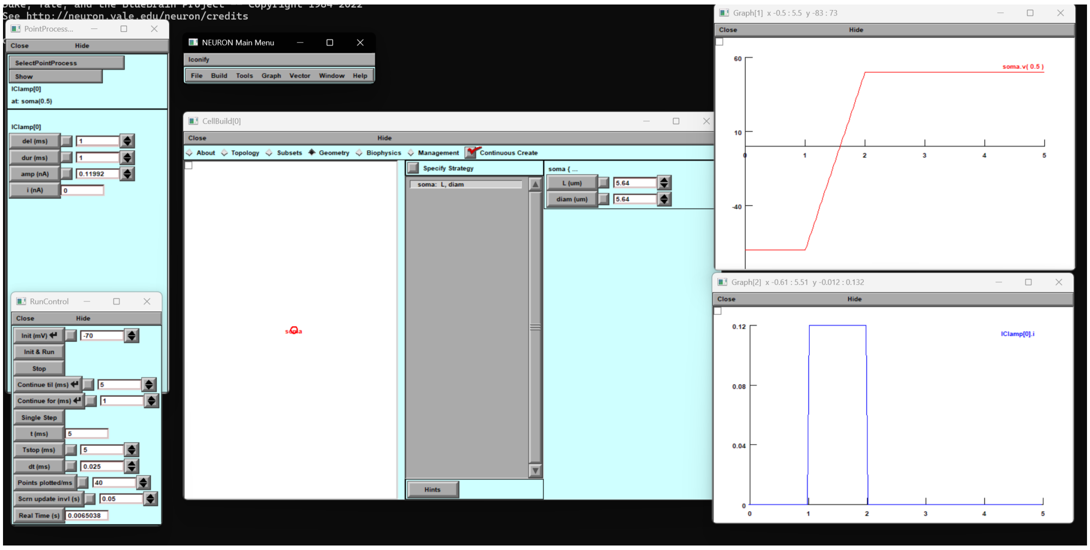

Figure 2: Question 1 Part (c)

Part (D)

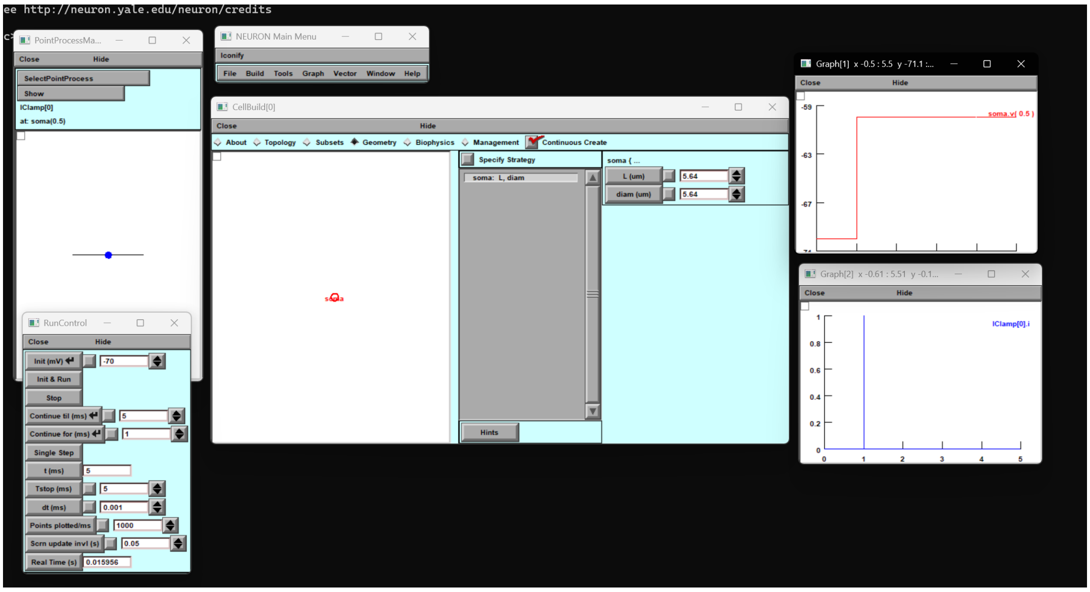

Figure 3: Question 1 Part (d)

Part (E)

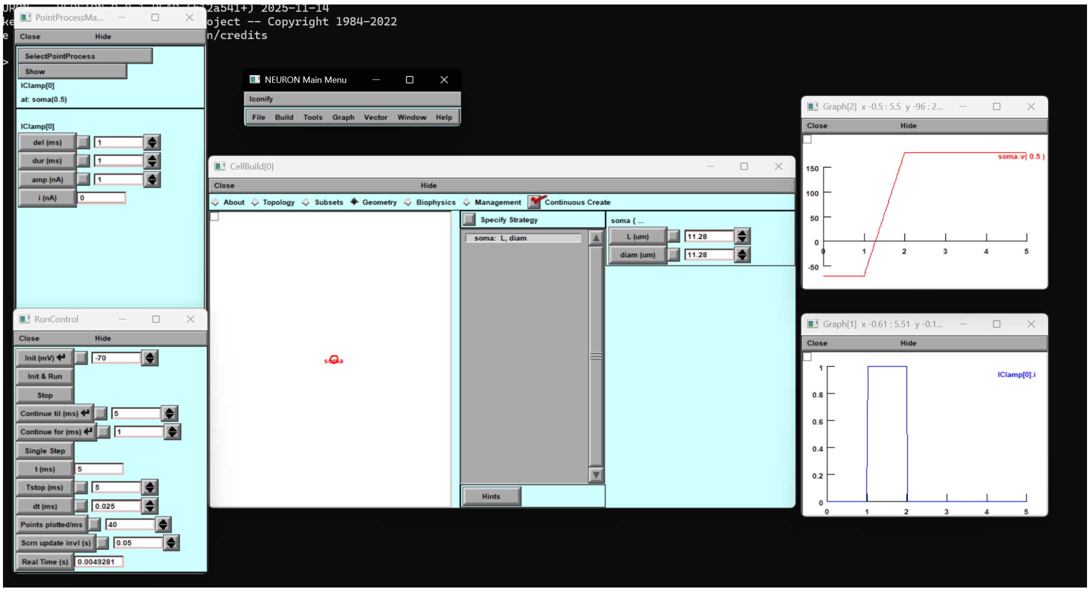

Figure 4: Question 1 Part (e)

Part (F)

$$C_A = C_m(\pi d^2) \approx 9.993 \times 10^{-13}\ \text{F} \approx 1.0\,\text{pF}.$$

Doubling diameter doubles d, and since $A = \pi d^2$, the membrane area becomes $4\times$. Because total capacitance is $C_{\text{tot}} = C_m A$, the total capacitance also becomes $4\times$:

$$C_B = 4C_A \approx 4.0\,\text{pF}.$$

Therefore, Neuron A has total capacitance $\approx 1.0\,\text{pF}$ and Neuron B has total capacitance $\approx 4.0\,\text{pF}$. This matches the intuition that a larger membrane area stores more charge for the same change in voltage.

Problem 2

Part (A)

Given passive conductance $g_{\text{pas}} = 0.001\,\text{S/cm}^2$ and $C_m = 1\,\mu\text{F/cm}^2$, the membrane time constant is

$$\tau = \frac{C_m}{g_{\text{pas}}} = \frac{1 \times 10^{-6}}{1 \times 10^{-3}} = 1 \times 10^{-3}\ \text{s} = 1\,\text{ms}.$$

Therefore, the membrane time constant is 1 ms.

Part (B)

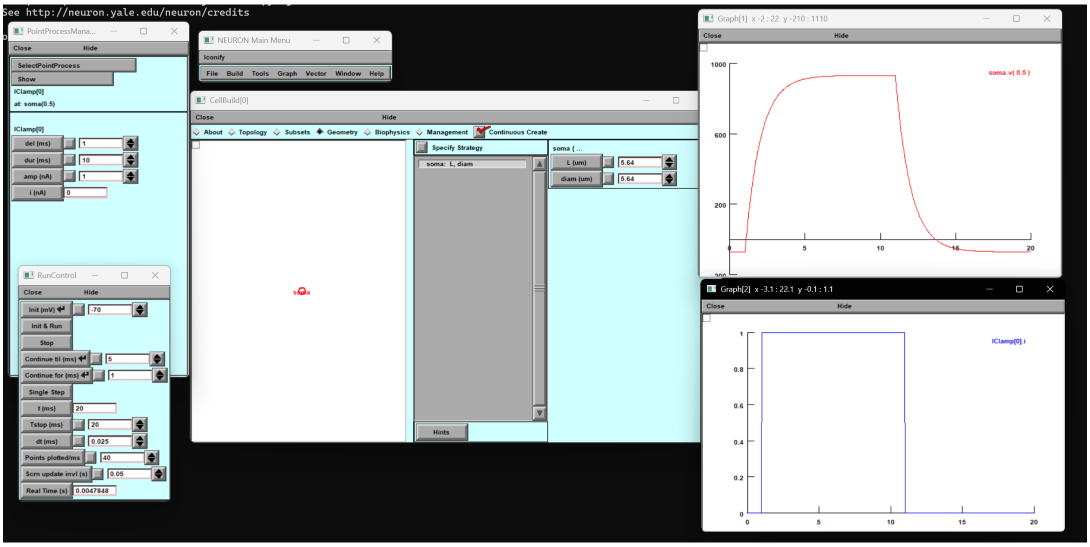

Figure 5: Question 2 Part (b)

Part (C)

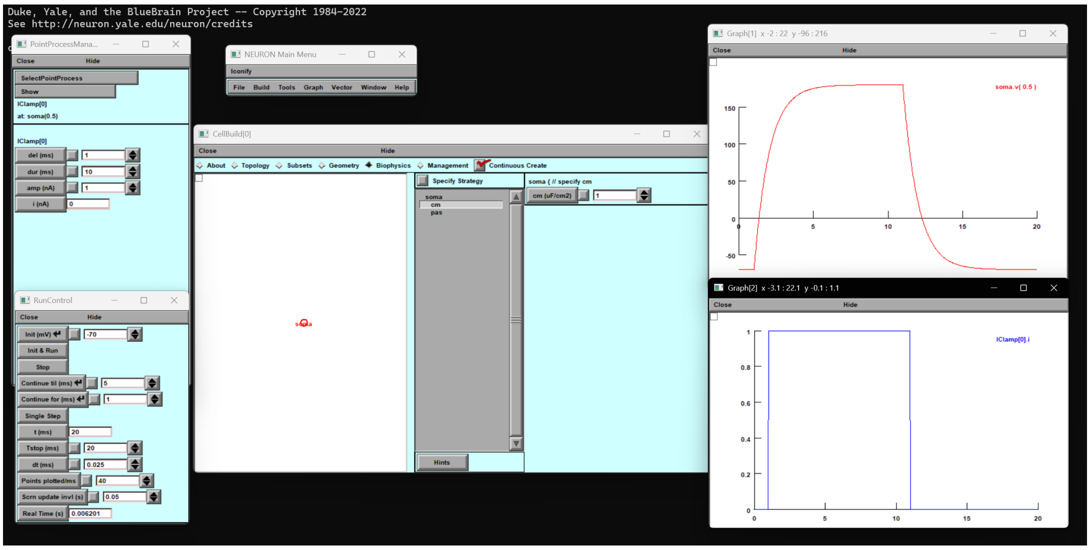

Figure 6: Question 2 Part (c)

Part (D)

Using the Neuron A surface area from Problem 1:

$$A_A \approx 9.993 \times 10^{-7}\ \text{cm}^2.$$

Total conductance $G_A = g_{\text{pas}} A_A$, so total resistance is

$$R_A = \frac{1}{G_A} = \frac{1}{g_{\text{pas}} A_A} = \frac{1}{(0.001)(9.993 \times 10^{-7})} \approx 1.00 \times 10^9\ \Omega.$$

For Neuron B (diameter doubled), area becomes $4A_A$, hence resistance becomes one-fourth:

$$R_B = \frac{1}{g_{\text{pas}}(4A_A)} = \frac{R_A}{4} \approx 2.50 \times 10^8\ \Omega.$$

Therefore, $R_A \approx 1.0\,\text{G}\Omega$ and $R_B \approx 250\,\text{M}\Omega$.

Part (E)

Membrane time constant $\tau = RC$. Since $C_{\text{tot}} = C_m A$ and $R_{\text{tot}} = \frac{1}{g_{\text{pas}} A}$,

$$\tau = \left(\frac{1}{g_{\text{pas}} A}\right)(C_m A) = \frac{C_m}{g_{\text{pas}}} = 1\,\text{ms}.$$

Therefore, Neuron A has membrane time constant $\tau_A = 1\,\text{ms}$ and Neuron B has membrane time constant $\tau_B = 1\,\text{ms}$.

Part (F)

Using the 63% method, the time constant is estimated from the rising phase as:

$$V(t_0 + \tau) = V_0 + 0.632\,(V_\infty - V_0).$$

For Neuron A, the step starts at $t_0 = 1\,\text{ms}$ with $V_0 = -70\,\text{mV}$ and reaches $V_\infty = 930.621\,\text{mV}$. The 63% level is $V_{63} \approx 562.086\,\text{mV}$, which is reached at $t_{63} = 2\,\text{ms}$. Thus,

$$\tau_A \approx t_{63} - t_0 = 2 - 1 = 1\,\text{ms}.$$

For Neuron B, the step starts at $t_0 = 1\,\text{ms}$ with $V_0 = -70\,\text{mV}$ and reaches $V_\infty = 180.155\,\text{mV}$. The 63% level is $V_{63} \approx 88.0215\,\text{mV}$, which is reached at $t_{63} = 2\,\text{ms}$. Thus,

$$\tau_B \approx t_{63} - t_0 = 2 - 1 = 1\,\text{ms}.$$

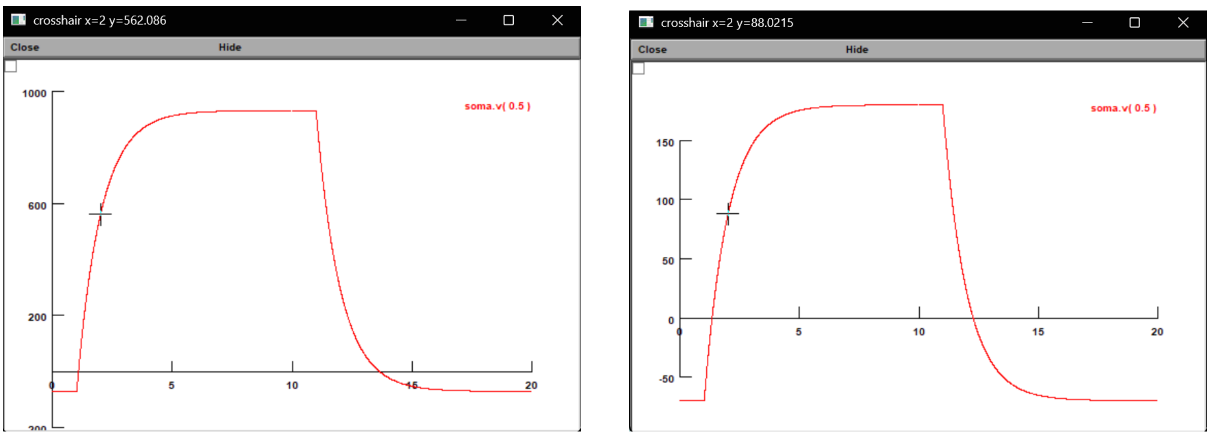

Figure 7: Question 2 Part (f): Neuron A (left) and Neuron B (right)

Problem 3

Part (A)

An axon with H&H channels was simulated with $L = 800\,\mu\text{m}$, $diam = 1\,\mu\text{m}$, and $R_a = 160\,\Omega \cdot \text{cm}$. A current pulse of amplitude $0.2\,\text{nA}$ and duration $1\,\text{ms}$ was injected at one end of the axon. The membrane potential was plotted at both ends and the center.

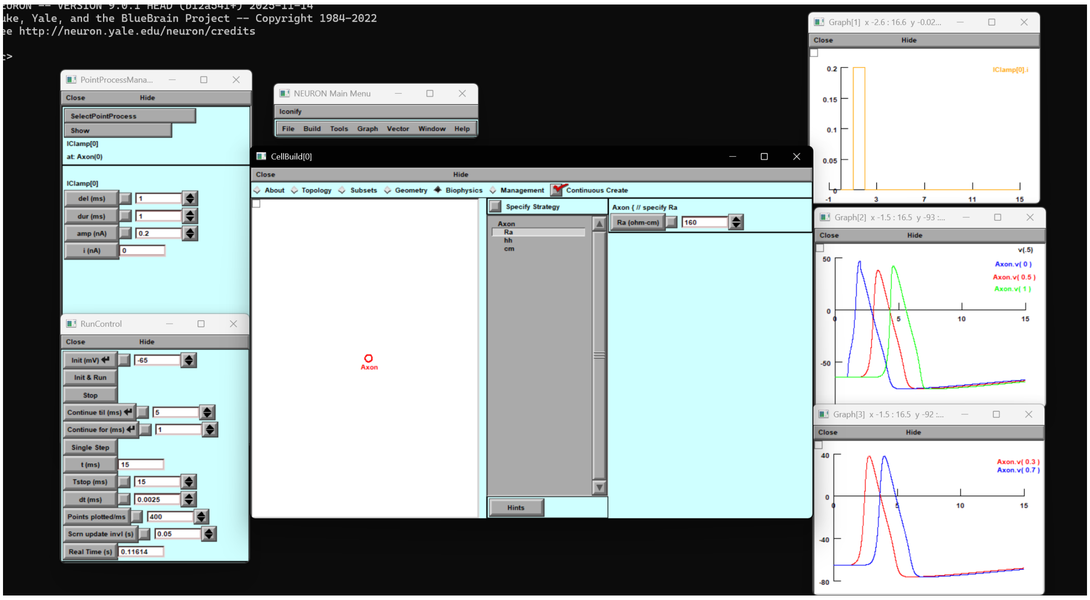

Figure 8: Question 3 Part (a)

Part (B)

To estimate conduction velocity, two locations were used: $x_A = 0.3L$ and $x_B = 0.7L$.

$$\Delta x = (0.7 - 0.3)L = 0.4 \times 800 = 320\ \mu\text{m}.$$

From the plot (peak times):

$$t_{0.3} = 2.785\,\text{ms}, \qquad t_{0.7} = 3.975\,\text{ms}.$$

$$\Delta t = t_{0.7} - t_{0.3} = 3.975 - 2.785 = 1.190\,\text{ms}.$$

Therefore,

$$v = \frac{\Delta x}{\Delta t} = \frac{320}{1.190}\ \mu\text{m/ms} \approx 268.9\,\mu\text{m/ms}.$$

Thus, the conduction velocity for the baseline axon is $\approx 268.9\,\mu\text{m/ms}$.

Part (C)

The axon diameter was doubled and the simulation was rerun.

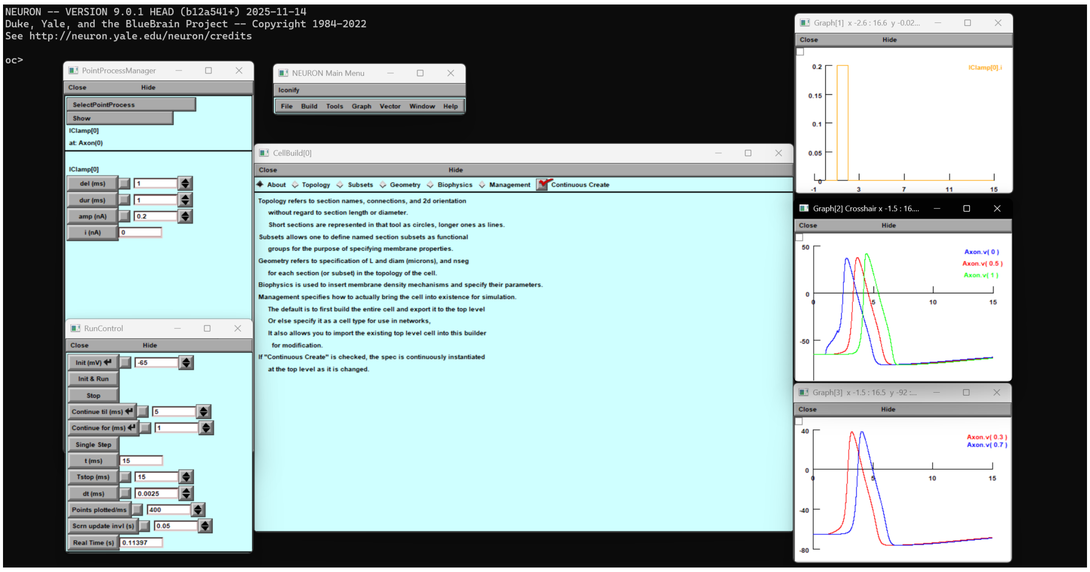

Figure 9: Question 3 Part (c)

Using the same points $0.3L$ and $0.7L$ (peak times):

$$t_{0.3} = 3.2425\,\text{ms}, \qquad t_{0.7} = 4.075\,\text{ms}, \qquad \Delta t_c = 4.075 - 3.2425 = 0.8325\,\text{ms}.$$

$$v_c = \frac{320}{0.8325}\ \mu\text{m/ms} \approx 384.4\,\mu\text{m/ms}.$$

Thus, after doubling diameter, the conduction velocity is $\approx 384.4\,\mu\text{m/ms}$.

Part (D)

The axon axial resistance R_a was doubled and the simulation was rerun.

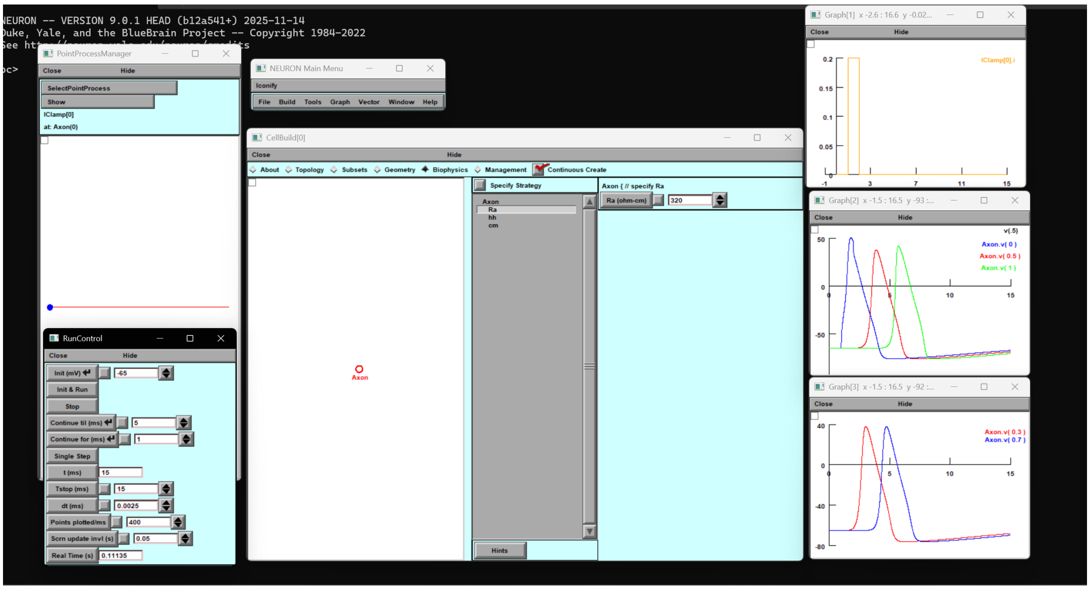

Figure 10: Question 3 Part (d)

Using the same points $0.3L$ and $0.7L$ (peak times):

$$t_{0.3} = 3.0425\,\text{ms}, \qquad t_{0.7} = 4.735\,\text{ms}, \qquad \Delta t_d = 4.735 - 3.0425 = 1.6925\,\text{ms}.$$

$$v_d = \frac{320}{1.6925}\ \mu\text{m/ms} \approx 189.1\ \mu\text{m/ms}.$$

Thus, after doubling axial resistance, the conduction velocity is $\approx 189.1\ \mu\text{m/ms}$.

Part (E)

Two current pulses were injected simultaneously at both ends of the axon.

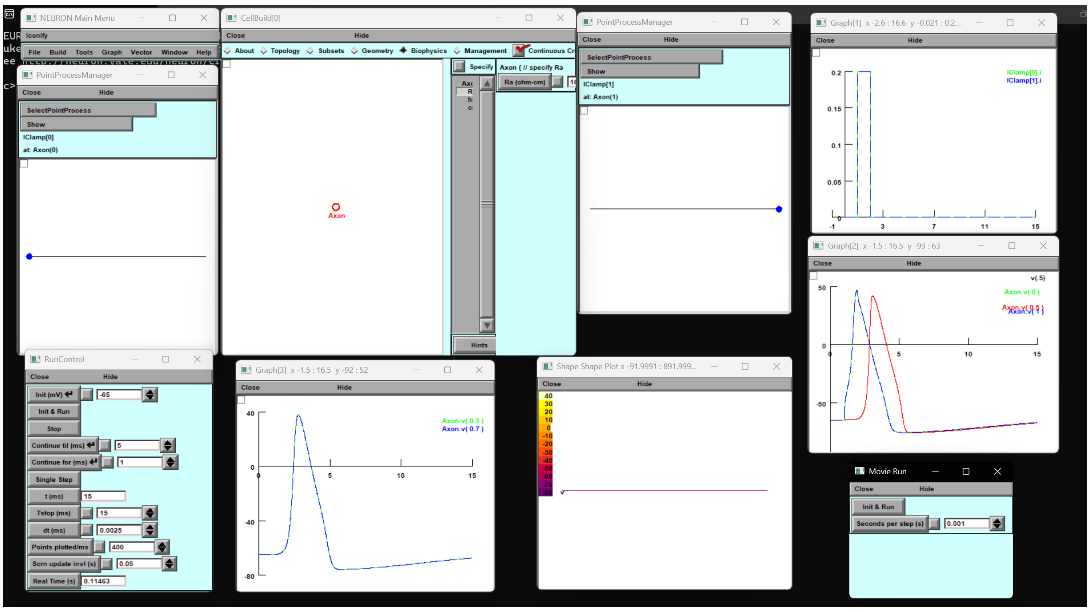

Figure 11: Question 3 Part (e)

In the Shape Plot, the color changed from purple to yellow starting from both corners (both ends of the axon) and moved inward, indicating two spikes propagating toward the center. When the waves approached the middle, they met and the axon then gradually returned back to purple along the length, showing repolarization and recovery toward resting state. In the voltage traces, axon.v(0.3) and axon.v(0.7) coincide, and axon.v(0) and axon.v(1) also coincide, which is expected due to symmetry (equal stimulation from both ends), so corresponding points are activated at the same time. The two spikes do not pass through each other: after collision the membrane in the meeting region is in a refractory state (recently activated sodium channels inactivated and potassium channels elevated), so it cannot immediately support a second spike, leading to apparent annihilation at the collision point.

Problem 4

Part (A)

A ball-and-stick model was built with:

Soma: $diam = 20\,\mu\text{m}$, $L = 20\,\mu\text{m}$, $nseg = 1$; Axon: $L = 1000\,\mu\text{m}$, $diam = 5\,\mu\text{m}$, $nseg = 101$.

Biophysics used:

Soma: $R_a = 160\,\Omega \cdot \text{cm}$, HH channels (default); Axon: $R_a = 160\,\Omega \cdot \text{cm}$, $C_m = 1$, pas with $\tau = 10\,\text{ms}$.

Initialization was set to $V_{init} = -65\,\text{mV}$. A stimulus current was injected at `soma(0.5)` with delay 1 ms, duration 1 ms, and amplitude 5 nA (chosen to reliably evoke a soma spike).

From the time plot, the soma shows a clear action potential because it contains HH channels. In the axon, since only passive channels are present, the response is mainly electrotonic spread (a graded depolarization) and does not behave like a regenerative traveling spike along the entire axon. The shape plot supports this: the depolarization spreads from the soma into the axon but attenuates with distance because the passive axon has no voltage-gated sodium channels to re-amplify the waveform.

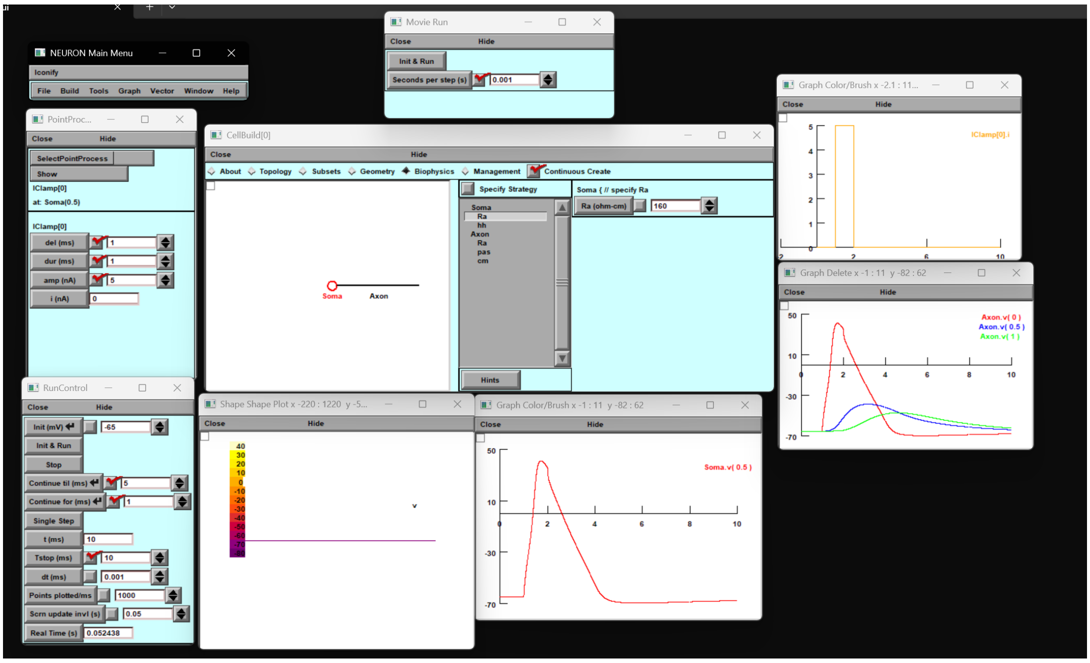

Figure 12: Question 4 Part (a)

Part (B)

In this part, HH channels were added to the axon as well (all other geometry, R_a, C_m, V_{init}, and the same current clamp settings were kept unchanged).

This model is different because the axon is now *active*: once the soma spike enters the axon, voltage-gated sodium and potassium channels in the axon can regenerate the spike. As a result, instead of only a decaying passive response, a proper action potential propagates along the axon with a clear delay between positions (and a more uniform spike shape compared to the passive case). The time plot and shape plot show a more distinct traveling wave along the axon compared to Part (A).

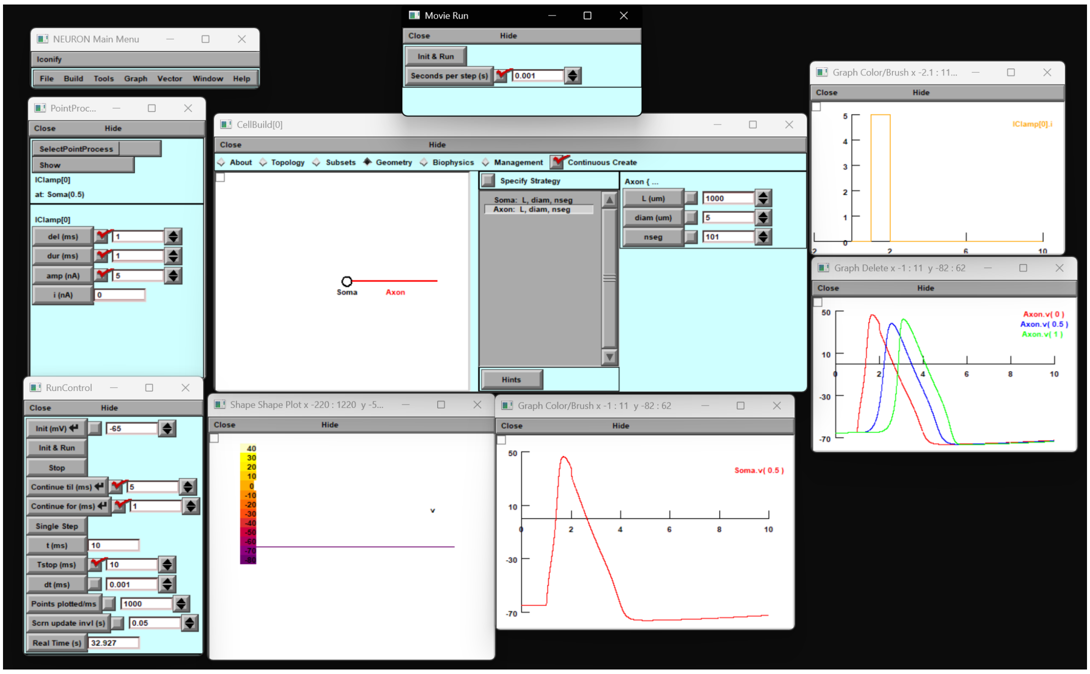

Figure 13: Question 4 Part (b)